Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 2 complete test ICs for all35 variants

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**0**	0.128	0.396	1.55	0.821	0.259	4.49	0.229	0.613	0.899	0	18/18
1 ANN10	**1**	0.156	0.564	4.38	0.781	0.592	1.66	0.2	4.1	333	0	17/18
1 ANN10	**10**	0.271	0.606	6.23	4.74	1.51	5.35	0.308	122	1.27e+03	0	14/18
1 ANN10	**100**	0.682	1.03	1	1	3.38	24.7	1	1.08	104	0	18/18
1 ANN10	**1000**	0.692	1.87	1	65.1	11	3.72	1	0.999	3.67e+03	0	18/18
2 P-only DeepONet10	**0**	0.154	0.361	4.64	0.921	0.305	3.03	0.183	1.03	1.01	0	18/18
2 P-only DeepONet10	**1**	0.132	0.523	0.854	2.72	0.893	1.99	0.238	4.58	2.09	0	18/18
2 P-only DeepONet10	**10**	0.418	0.996	1.91	1	11.9	3.15	0.84	253	8.52e+03	0	18/18
2 P-only DeepONet10	**100**	0.606	1.77	1	1	7.05	13.2	1	1.09	2.42	0	6/18
2 P-only DeepONet10	**1000**	0.581	1.56	1	1	6.75	1	1	2.01e+04	1	0	18/18
3 ANN10 + P-only DeepONet10	**0**	0.0981	0.388	0.906	0.537	0.33	2.68	0.159	0.582	0.585	0	18/18
3 ANN10 + P-only DeepONet10	**1**	0.127	0.499	0.857	4.72	0.622	0.956	0.204	19	2.9	0	18/18
3 ANN10 + P-only DeepONet10	**10**	0.274	0.69	3.25	3.84	1.31	6.11	0.604	1.71e+03	2.76e+05	0	18/18
3 ANN10 + P-only DeepONet10	**100**	0.689	0.887	1	1	3.77	25	1	1.07	35.1	0	18/18
3 ANN10 + P-only DeepONet10	**1000**	0.569	1.94	1	1	12.5	1	1	0.998	5.05e+03	0	18/18
4 T/P DeepONet10	**0**	0.1	0.401	2.38	0.477	0.372	0.566	0.153	5.65	410	0	17/18
4 T/P DeepONet10	**1**	0.184	0.555	2.58	1.82	2.31	1.06	0.272	552	1.33e+04	0	9/18
4 T/P DeepONet10	**10**	0.64	0.951	31.6	13.6	6.04	7.43	0.754	185	8.49e+03	0	15/18
4 T/P DeepONet10	**100**	0.709	1.01	1.41	17.5	2.87	21.2	0.984	2.73	2.09	0	18/18
4 T/P DeepONet10	**1000**	0.582	1.54	1	1	5.35	1	1	2.15e+04	1	0	18/18
5 ANN10 + T/P DeepONet10	**0**	0.4	0.491	29	0.581	0.566	0.462	0.372	111	1.87e+03	0	18/18
5 ANN10 + T/P DeepONet10	**1**	0.126	0.49	1.13	0.425	0.464	0.321	0.188	0.697	0.931	0	18/18
5 ANN10 + T/P DeepONet10	**10**	0.194	0.546	4.29	2.78	0.843	3.24	0.261	95.1	234	0	18/18
5 ANN10 + T/P DeepONet10	**100**	0.474	1.58	1.24	8.21	6.05	1	0.772	1.48e+03	5.24e+04	0	18/18
5 ANN10 + T/P DeepONet10	**1000**	0.569	1.94	1	1	12.5	1	1	0.997	4.67e+03	0	18/18
6 T/P/ϕ DeepONet10	**0**	0.374	1.32	1.34	1.24	1.04	2.14	0.553	1.2	3.79	0	17/18
6 T/P/ϕ DeepONet10	**1**	0.45	0.839	0.962	69.1	3.54	2.28	0.535	8.71	0.92	0	11/18
6 T/P/ϕ DeepONet10	**10**	0.61	1.04	7.14	2.58	5.87	1.59	0.744	67.1	6.28e+03	0	17/18
6 T/P/ϕ DeepONet10	**100**	0.727	0.826	8.51	7.96	3.65	18.4	0.889	3.96	2.67	0	16/18
6 T/P/ϕ DeepONet10	**1000**	0.572	1.96	1	1	12.6	1	1	3.78	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.305	1.31	0.575	0.586	0.986	0.598	0.505	0.714	0.67	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.297	0.509	2.06	4.06	0.421	5.64	0.355	0.965	1.45	0	17/18
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.169	0.541	2.66	2.65	0.812	2.53	0.244	114	210	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.542	0.936	1.32	8.52	6.69	9.43	0.791	925	9.23e+03	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.567	1.93	1	1	12.4	1	1	0.996	6.41e+03	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 44, 53.